LLMs for 6G Networks: Fine-Grained Telecom Traffic Generation and Forecasting

From Coarse Operator Measurements to Risk-Aware Traffic Prediction

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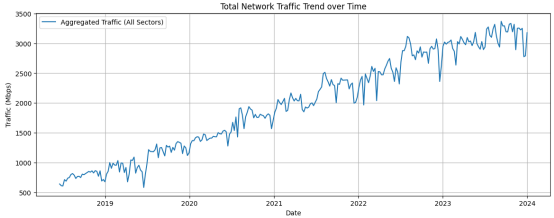
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Why do we need more than point forecasting?

- Mobile networks are often provisioned for busy periods.
- Sleep modes can save energy during low load.
- A mean forecast is not enough for safe control.
- Short unexpected bursts can still create delay and QoS degradation.
- We therefore need probabilistic forecasting, especially upper quantiles.

The Bouygues dataset is useful, but too coarse

- 86 sectors across 29 sites
- Coarse traffic measurements, treated as 5-minute averages
- Good for macro trends
- Not sufficient for instantaneous burst modeling
- Not sufficient to directly train fine-grained forecasting models



KTH IPP framework: from coarse averages to bursty traffic

- ON/OFF continuous-time Markov chain
- Poisson arrivals during ON
- Exponential traffic marks
- Goal: generate fine-grained traffic consistent with coarse slot statistics

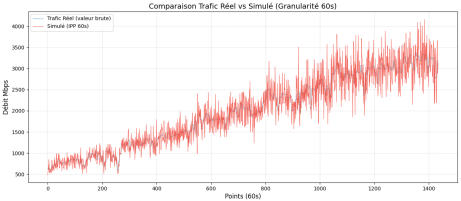
$$P_{\mathrm{ON}} = \frac{\tau}{\tau + \zeta} \qquad r = P_{\mathrm{ON}} \lambda \psi$$

Why did we need a variance proxy?

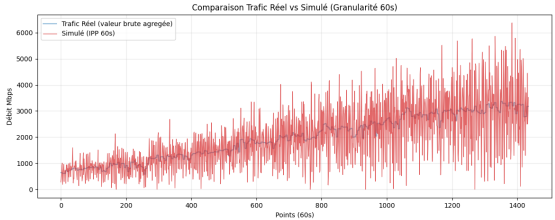
- Original KTH needs a mean and a variance for each slot
- Bouygues directly gives the mean, but not the variance in the required form
- We estimated a local variance from the coarse signal itself
- Then we stabilized it with two floors:
  - generic floor
  - KTH feasibility floor

$$\texttt{coarse signal} \rightarrow \texttt{rolling local variance} \rightarrow \texttt{floors} \rightarrow \texttt{final variance proxy}$$

Aggregation was the turning point



example target



Our aggregated simulated vs real 60s signal

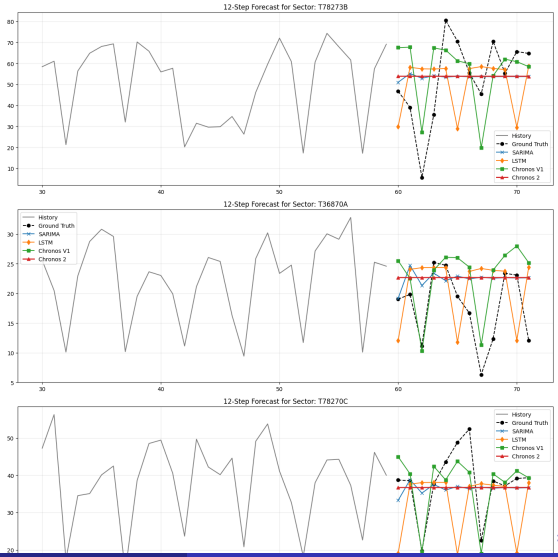
- Sector-level generated traffic looked too sparse
- Aggregated traffic across sectors looked realistic
- Macro-scale comparison matched the expected KTH-style behavior
- Law of large numbers smooths extreme micro-bursts

Forecasting benchmark

- SARIMA: classical baseline
- LSTM: trained recurrent model
- Chronos V1: zero-shot foundation model
- Chronos 2: newer zero-shot foundation model
- Context length = 60
- Horizons = 12-step and 1-step
- Metrics = RMSE and pinball loss

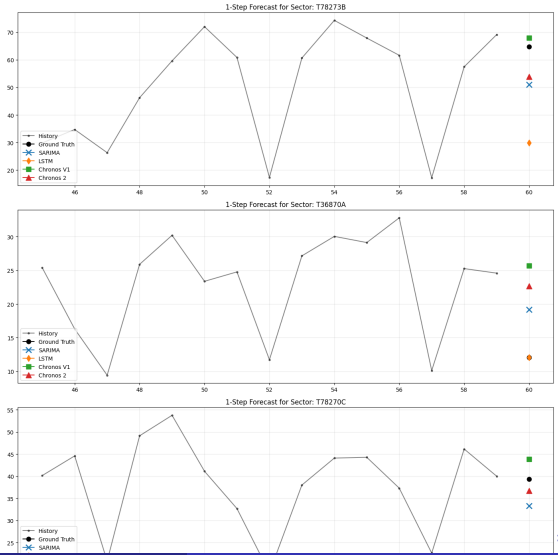
12-step forecasting results

Model	RMSE	Pinball
SARIMA	15.57	6.30
Chronos 2	15.74	6.39
Chronos V1	16.16	6.52
LSTM	17.89	7.50



1-step forecasting results

Model	RMSE	Pinball
Chronos 2	11.51	5.76
SARIMA	12.98	6.49
Chronos V1	14.99	7.50
LSTM	27.72	13.86



Main takeaways

- We started from coarse industrial telecom data
- We adapted the KTH IPP framework to generate fine-grained traffic
- The rolling variance proxy made second-moment matching possible
- Aggregated macro validation confirmed the realism of the generated signal
- Chronos 2 gave the best single-step benchmark
- Next step: evaluate upper quantiles such as 0.9 or 0.95 for truly risk-aware forecasting